S1.2 The nuclear atom

The big picture

Guiding question(s)

How do the nuclei of atoms differ?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

For many people, nuclear power is a topic with negative connotations. They tend to think about the negative consequences of nuclear power such as the disposal of nuclear waste or the various nuclear accidents that have occurred throughout history. Here we will focus on the positive effects of the nuclear industry, specifically the use of radioisotopes. Radioisotopes are radioactive isotopes that are unstable because of the numbers of subatomic particles they contain in their nuclei. Because of their instability, they emit various forms of radiation that can be used for different purposes, some of which are listed below.

- Carbon-14 is used for carbon dating, a technique that can be used to determine the age of an object that contains organic material.
- Cobalt-60 is used for killing bacteria on food, in a technique known as food irradiation.
- Americium-241 is used in smoke detectors which can detect house fires.
- Technetium-99 is used for medical diagnosis and cancer treatment.
- Hydrogen-3 is used as a radioactive tracer to provide evidence for a specific reaction mechanism.

While sometimes controversial, these radioisotopes play important roles in many different industries from food production to medical diagnosis. How do you feel about the use of such radioisotopes? Do the benefits outweigh the risks?

Concept

The structure of an isotope can make it unstable because of the number of subatomic particles it contains in its nucleus.

Figure 1. Apples being irradiated with the radioisotope cobalt-60.

Source: "Cobalt-60 Irradiator.tif (https://commons.wikimedia.org/wiki/File:Cobalt-60_Irradiator.tif)" by pnl.gov, US Department of Energy is in the public domain

Making connections

Subtopic R3.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/) covers how isotope tracers can be used to provide evidence for a specific reaction mechanism.

Prior learning

Before you study this subtopic, keep in mind that elements:

- Are pure substances (see section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/))
- Are composed of only one type of atom (see section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/))